EXERCISE 05

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1 Introduction

Solution of Exercise 05.

2 Methodology

2.1 Parametric and iterative solutions

Writing $I(R)$ for the integral in the quenched equation, choose $t = R$, with $0 \le R < 1$. The curve is then described by

$$\boxed{\alpha(R) = \frac{\pi R}{\sqrt{1-R}\, I(R)}, \qquad \epsilon(R) = \frac{\arccos R}{\pi}.} \tag{1}$$

For fixed $\alpha > 0$, the same equation admits the two rearrangements

$$f_1(R,\alpha) = \frac{\alpha}{\pi}\sqrt{1-R}\, I(R), \qquad f_2(R,\alpha) = 1 - \frac{\pi^2 R^2}{\alpha^2 I(R)^2}. \tag{2}$$

Both give the physical fixed point R^*, and are iterated via

$$R_{i+1} = T_j(R_i;\alpha,a) = (1-a)R_i + a f_j(R_i,\alpha), \qquad 0 < a \le 1. \tag{3}$$

The integral is computed by adaptive quadrature (10^{-10} absolute and relative tolerances). To avoid cancellation and underflow in the Gaussian tail, we evaluate $\log H(u) = \log_\mathrm{ndtr}(-u)$ before exponentiating the integrand. Iteration terminates when $|R - f_j(R,\alpha)| < 10^{-10}$, or after 10^3 updates; non-finite iterates and $R \notin [0,1)$ are rejected.

2.2 Local convergence and asymptotic analysis

Let R^* solve $R^* = f_j(R^*, \alpha)$. Linearisation gives

$$R_{i+1} - R^* = \lambda_j (R_i - R^*) + O\big((R_i - R^*)^2\big), \quad \lambda_j = 1 - a + a f_j'(R^*, \alpha). \tag{4}$$

Thus $|\lambda_j| < 1$ ensures *local* convergence. Differentiating (2) and using the fixed-point equation yields

$$f_1'(R^*, \alpha) = -\frac{R^*}{2(1-R^*)} + \frac{R^* I'(R^*)}{I(R^*)}, \tag{5}$$

$$f_2'(R^*, \alpha) = -\frac{2(1-R^*)}{R^*} \left(1 - \frac{R^* I'(R^*)}{I(R^*)}\right). \tag{6}$$

For completeness, with $\phi(v) = e^{-v^2/2}/\sqrt{2\pi}$ and $H'(u) = -\phi(u)$,

$$I'(R) = \int_{-\infty}^{\infty} dv \, \frac{\phi(v) e^{-Rv^2/2}}{H(-\sqrt{R}\, v)} \left[-\frac{v^2}{2} - \frac{v\, \phi(\sqrt{R}\, v)}{2\sqrt{R}\, H(-\sqrt{R}\, v)} \right]. \tag{7}$$

Expanding $H(-x) = \frac{1}{2} + x/\sqrt{2\pi} + O(x^3)$ near $x = 0$ gives $I(0) = 2$ and $I'(0) = 4/\pi - 1$. Hence, as $\alpha \to 0^+$, $R^* \sim 2\alpha/\pi$, so $f_1'(R^*) \to 0$ whereas $f_2'(R^*) \sim -2/R^* \to -\infty$. Consequently $\lambda_1 \to 1 - a$, but f_2 loses stability for any fixed $a > 0$.

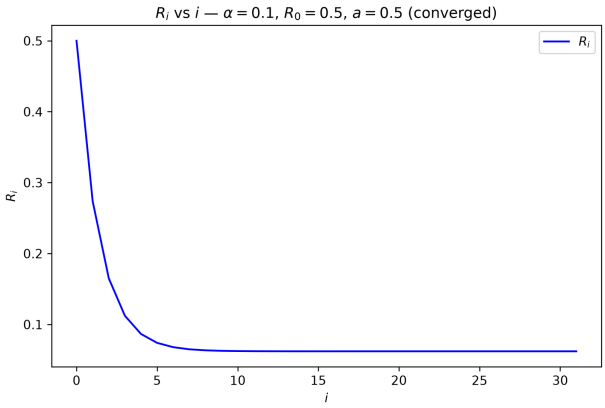
In the opposite limit, $R^* \to 1^-$, while $I(1)$ and $I'(1)$ remain finite (the integrands have Gaussian tails). Thus $f_1'(R^*) \to -\infty$ and $f_2'(R^*) \to 0$: now $\lambda_2 \to 1 - a$ and f_1 becomes unstable. *Neither rearrangement is suitable throughout $\alpha \in (0, \infty)$ with a fixed positive interpolation parameter.* As a further check, these limits predict

$$\epsilon(\alpha) = \frac{1}{2} - \frac{2\alpha}{\pi^2} + O(\alpha^2) \quad (\alpha \to 0^+), \qquad \epsilon(\alpha) \sim \frac{\sqrt{2}}{\alpha I(1)} \quad (\alpha \to \infty), \quad I(1) \simeq 2.264. \tag{8}$$

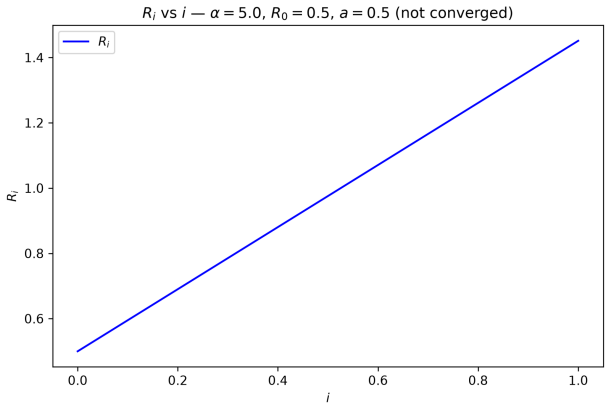
3 Results

3.1 Iteration traces

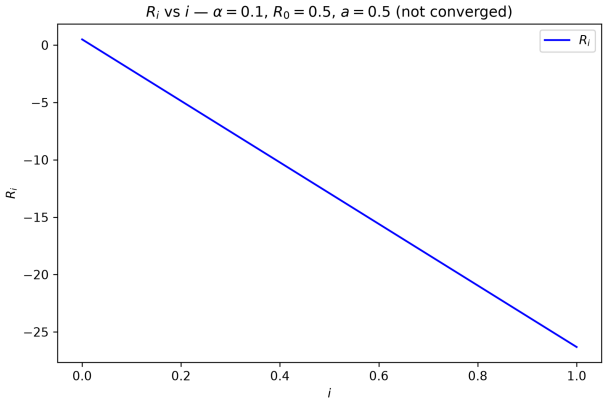
For the prescribed $R_0 = a = 0.5$, f_1 converges at $\alpha = 0.1$ in 31 updates ($R^* \simeq 0.06217$), while f_2 converges at $\alpha = 5$ in 27 updates ($R^* \simeq 0.93206$). The opposite pairings exit the physical interval on the *first* update. Their two-point plots therefore represent failed steps.



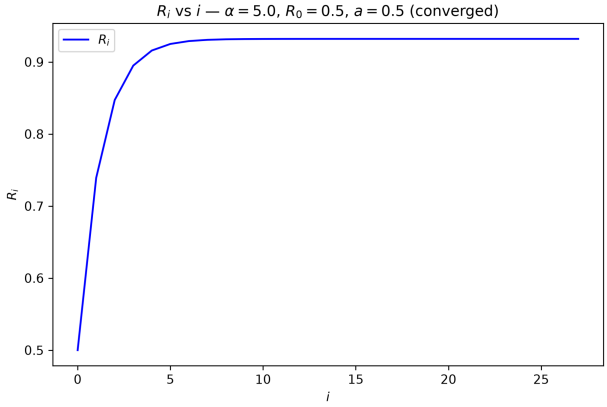
(a) f_1, $\alpha = 0.1$: converged.



(b) f_1, $\alpha = 5$: invalid first update.



(c) f_2, $\alpha = 0.1$: invalid first update.



(d) f_2, $\alpha = 5$: converged.

Figure 1: Iterations R_i against i for both maps, with $R_0 = 0.5$ and $a = 0.5$.

3.2 Convergence survey and automatic procedure

We test 100 uniformly spaced $\alpha \in [0, 10]$, 10 initial overlaps $R_0 \in [0.01, 0.99]$, and 10 interpolation parameter values $a \in [0.1, 0.9]$ for each map. The heatmaps show the fraction of successful α values for each (R_0, a); f_1 is mainly successful for smaller a, whereas f_2 is robust over a much wider range.
Neither map achieves a 100% success rate over the entire sampled grid: f_1 becomes unstable at large α, while f_2 fails at sufficiently small α and is undefined at $\alpha = 0$.

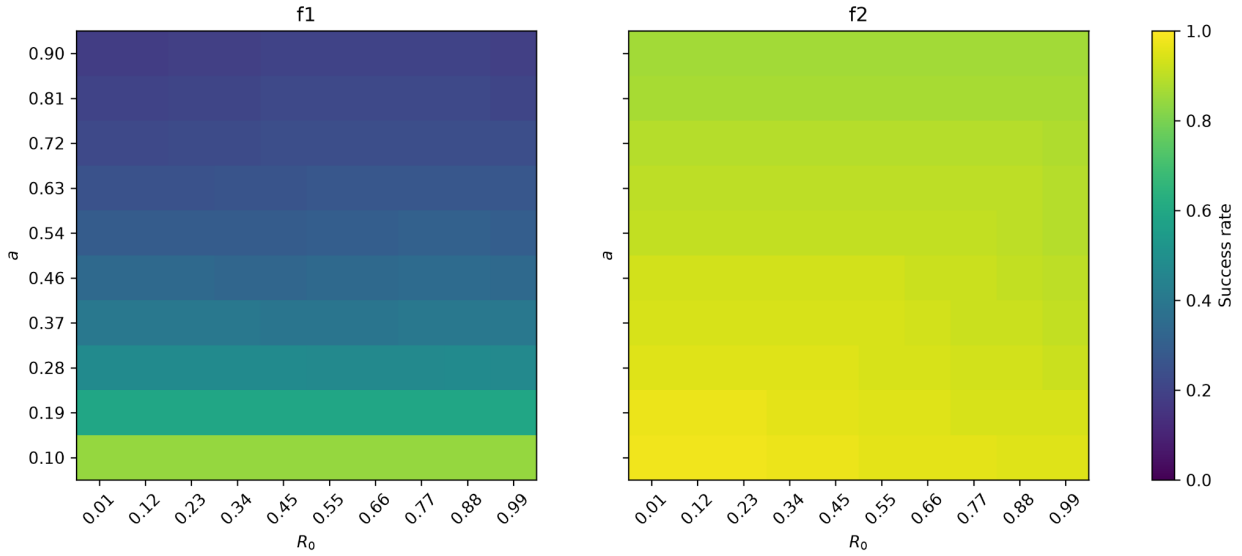


Figure 2: Success fraction over the α grid as a function of (R_0, a), separately for f_1 and f_2.

The automatic solver takes the midpoint of the sampled interval in which *both* maps have a 100% success rate, obtaining $\alpha_{\text{switch}} = 1.565657$. For each side of the threshold, it retains configurations with 100% success and ranks them by mean residual and then mean iteration count. The selected settings are

range	map	(R_0, a)
$\alpha \leq 1.565657$	f_1	$(0.663333,\ 0.722222)$
$\alpha > 1.565657$	f_2	$(0.445556,\ 0.900000)$

(9)

At $\alpha = 0$ the code sets $R = 0$ and $\epsilon = 1/2$ directly; at every positive grid point, it restarts from the selected R_0 and enforces the same 10^{-10} residual tolerance.

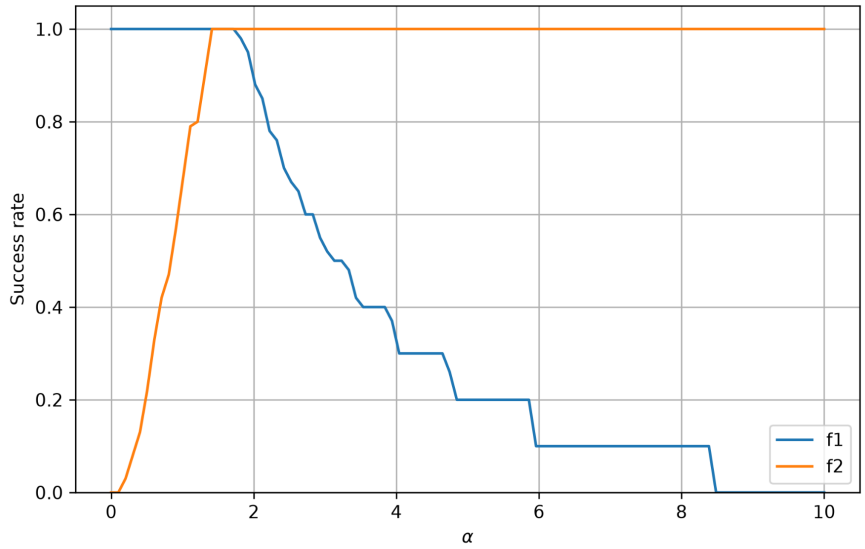


Figure 3: Success rate of both maps versus α.

3.3 Final learning curve

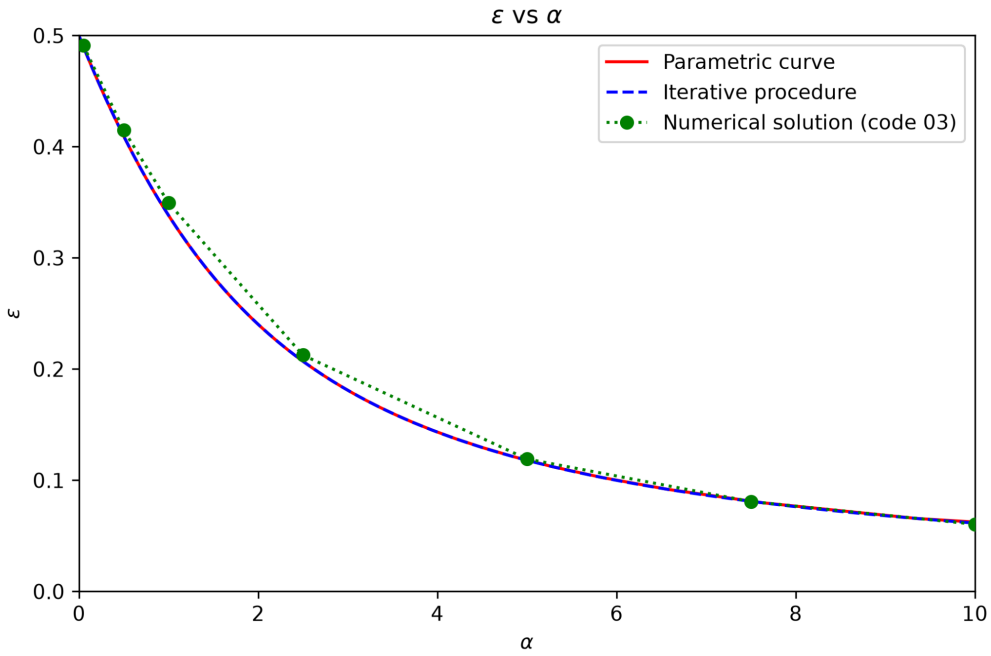


Figure 4: Quenched generalisation curve obtained from the parametric and iterative solutions, compared with the Exercise 03 simulation at $N = 20$.

The automatic and parametric curves are indistinguishable at the plotted scale, confirming the iterative solution. The Exercise 03 points are also close to the quenched prediction, with small finite-size and sampling discrepancies.

4 Conclusions

The linear stability analysis explains why f_1 works best for small α and f_2 for large α. Switching between them yields a high-precision quenched learning curve on $0 \leq \alpha \leq 10$, consistent with both the parametric construction and the finite-N numerical experiment.